Understand Inequalities and Their Solutions

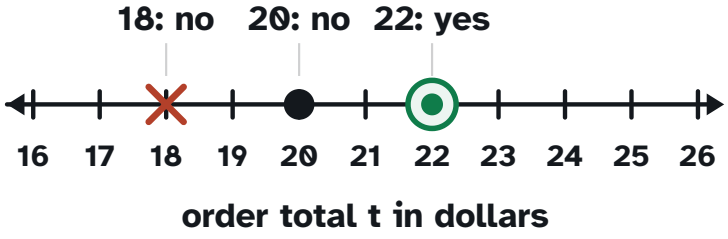
Test numbers in an inequality to see which ones are solutions.

Name _____ **Date** _____

Tonight's spark

A pizza coupon says: "Order total over $20 gets a free drink." Does an order of exactly $20 get the free drink? What about $21?

Guess first: does exactly $20 get a free drink, yes or no?



Our class model: test one value, then decide

Try it

Work it out on paper. Write one answer on each line.

1 **Is 5 a solution of x > 3?**

A. Yes, because 5 > 3 is true **B.** No, because 5 > 3 is false

C. No, because 5 does not equal 3 **D.** Only 3 is a solution

My answer _____

2 **Is 4 a solution of x ≥ 4?**

A. No, because 4 is not more than 4 **B.** Yes, because 4 ≥ 4 is true **C.** Yes, because 4 > 4

D. No, because inequalities never allow equal

My answer _____

3 **Is 4 a solution of $x < 4$?**

A. Yes, because $4 = 4$ **B.** Yes, because $4 < 4$ **C.** No, because $4 < 4$ is false

D. No, because 4 is negative

My answer _____

4 **Which value is a solution of $x \leq 9$?**

A. 10 **B.** 11 **C.** 12 **D.** 9

My answer _____

5 **How many solutions does $x > 2$ have?**

A. Exactly one **B.** Exactly two **C.** Many **D.** None

My answer _____

Talk about it

Was the boundary number a solution? How does that tell you which circle to draw?

Home mission

Find a real coupon or receipt. Say: "Over $20 gets a free drink." Is $20 > 20$ true? Test $21 and $25 the same way. What does that tell you about the circle at 20?

If we get stuck, ask

- Put that number in. Is the statement true or false?
- What happens when you test the boundary number itself?
- Does the symbol point toward bigger numbers or smaller ones?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/8-5-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it